

December 2018

ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. A. Choose the most appropriate answer from the following (any ten): 1×10
- i) Which of the following material has maximum ductility? - (a) Mild steel, (b) Copper, (c) Nickel, (d) Aluminium.
 - ii) The hardness is the property of a material due to which it: - (a) can be drawn into wires, (b) breaks with little permanent distortion, (c) can cut another metal, (d) can be rolled or hammered into thin sheets.
 - iii) The percentage of carbon in cast iron varies from: - (a) 0.1 to 0.5, (b) 0.5 to 1, (c) 1 to 1.7, (d) 1.7 to 4.5.
 - iv) Rapid quenching of austenite in the heat treatment of steel produces a super saturated and distorted crystal structure which is known as: - (a) cementite, (b) pearlite, (c) martensite, (d) ferrite.
 - v) Which among the following is a para-magnetic material? - (a) Iron, (b) Cobalt, (c) Nickel, (d) Aluminium.
 - vi) Bronze is a alloy of: - (a) copper & zinc, (b) copper, tin & zinc, (c) copper & tin, (d) none of these.
 - vii) Dye penetrant method is generally used to locate- (a) core defects, (b) surface defects, (c) superficial defects, (d) temporary defects.
 - viii) Silicon when added to copper improves: - (a) machinability, (b) hardness, (c) hardness and strength, (d) strength and ductility.
 - ix) The lower critical point for all steels is: - (a) 600°C, (b) 700°C, (c) 723°C, (d) 913°C.
 - x) The heat treatment process used for softening hardened steel is: - (a) carburizing, (b) normalizing, (c) annealing, (d) tempering.
 - xi) The hardness of steel increases if it contains: - (a) pearlite, (b) ferrite, (c) cementite, (d) martensite.
 - xii) Non-destructive testing is used to determine: - (a) location of defects, (b) chemical composition, (c) corrosion of metal, (d) All of these.
 - xiii) Which of the following tool is manufactured by powder metallurgy?: - (a) High speed steel, (b) Sintered carbides, (c) High carbon steel, (d) Low carbon steel.

B. State whether the following statements are true or false (any ten):

1×10

- i) Carbon is present in the form of graphite in gray cast iron.
- ii) Nickel increases strength and toughness in alloy steel.
- iii) Compression test is a type of non-destructive test.
- iv) Thermal conductivity of non metal is inversely proportional to temperature.
- v) Stiffness is the ability of a material to resist deformation under stress.
- vi) Pitting is a localized form of corrosion.
- vii) An eutectoid steel consists of pearlite and cementite.
- viii) Galvanization is a process of corrosion control of steel.
- ix) Chromium ranges from 16-18 percentages in ferritic stainless steel.
- x) Duralumin is a ferrous alloy.
- xi) Blast furnace is used to produce cast iron.
- xii) Steel with carbon above 0.8% is known as hyper-eutectoid steel.

- 2. a) State the differences between thermosetting and thermoplastic. Give two examples of each type.
b) What is annealing? What is the purpose of annealing process? (2+2)+(4+2)
- 3. a) What is powder metallurgy? Describe sintering process.
b) Differentiate between toughness and resilience.
c) List the applications of low carbon steel. (2+4)+2
- 4. a) What are composite materials? Explain with examples.
b) Distinguish between the full annealing and process annealing?
c) What is case hardening? 4+4+2
- 5. a) Why is cast iron used as a material for machine tool beds?
b) List the properties required for a material to withstand high temperature?
c) Give the composition of High speed steel? 3+5+2
- 6. a) Explain TTT curve for Eutectoid steel? <https://www.wbscteonline.com>
b) What do you mean by Tempering? What are different tempering processes? 5+5
- 7. a) Explain the corrosion mechanism?
b) What are the different methods of preventing corrosion? Briefly explain. 4+6
- 8. a) Name the various methods of manufacturing metal powders and give brief description of each.
b) What is infiltration? Give the metal powders used for porous bearings. 5+5
- 9. a) What do you mean by Engineering materials? Classify the various Engineering materials?
b) What is fatigue? How fatigue failure occurs? 7+3
- 10. a) Describe the types of rubber you know. Explain briefly.
b) Give some typical applications of Thermoplastic elastomers. 6+4

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December 2019

ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. Fill in the blanks (any twenty):

1 X 20

- i) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
- ii) Grey cast iron is _____ than white cast iron.
- iii) The iron ore mostly used for the production of pig iron is _____.
- iv) The plasticity is that property of a material by virtue of which it deforms _____ under the action of external force.
- v) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
- vi) The ultimate stress of a material is _____ than the breaking stress.
- vii) Cutting tools are made from _____ steel.
- viii) Grey cast iron is _____ than white cast iron.
- ix) The iron ore mostly used for the production of pig iron is _____.
- x) Ball bearing are usually, made from _____ steel.
- xi) An alloy of copper, tin and zinc is known as _____.
- xii) German silver consists of _____.
- xiii) Babbit metal is a _____ based alloy.
- xiv) The metal suitable for bearing subjected to high load is _____.
- xv) Brass is an alloy of _____ and _____.
- xvi) The heat treatment process used for softening hardened steel is _____.
- xvii) The process used for relieving the internal stresses previously set up in the metal and for increasing the machinability of steel is _____.
- xviii) Carburizing is a _____ process.

- xix) Normalizing _____ the mechanical properties of steel.
- xx) Normalizing of steel is done to _____ the grain structure.
- xxi) Phenolic resins are properly known as _____.
- xxii) The property of a material by virtue of which it can withstand or support an external force or load without rupture, is known as _____.
2. a) What do you mean by heat treatment of steel?
b) Describe any two of the following heat treatment processes – i) cyaniding, ii) carburising, iii) induction hardening, iv) nitriding. 4+6
3. a) Write a short note on corrosion.
b) What is the difference between creep and fatigue?
c) Write a short note on – i) pitting, ii) erosion. 3+3+4
4. a) Define alloy steel. Explain the effect of nickel, chromium and manganese on the properties of steel.
b) Describe aluminium casting alloys and Y-alloy in brief. 6+4
5. Describe: blending, compacting, pre-sintering and sintering method. 10
6. a) Write briefly about the different stages of powder metallurgy.
b) State the advantages of powder metallurgy. 8+2
7. a) How do thermo plastics differ from thermosetting plastics?
b) Write short notes on – i) Thermocole, ii) Glass wool.
c) State the properties of refractory materials. 3+4+3
8. a) Draw iron carbon equilibrium diagram and write the name of different phases and mention different.
b) What is the Duralumine? 7+3
9. a) State the properties and applications of copper alloy.
b) What is bearing material? 7+3
10. a) Difference between destructive testing and non destructive testing.
b) Write a short note on – i) Dry penetrant test, ii) Magnaflux test. 5+5

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MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. When a steel is heated in a furnace and then cooled in air at ordinary temperature, the process is
(a) Annealing (b) Hardening (c) Normalizing (d) Tempering
 - ii. γ form of iron is?
a) Magnetic b) Non magnetic c) Occurs below 768°C d) Occurs above 1410°C
 - iii. Bronze is an alloy of
(a) copper & zinc (b) copper & tin (c) copper, zinc & tin (d) any one of these
 - iv. Fatigue failure can be detected with the help of – a) stress – strain curve, b) S – N curve, c) graph of law of machine, d) iron – carbon diagram.
 - v. In low carbon steel C % varies from- (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - vi. In the austempering process of heat treatment, austenite changes into
a) martensite b) troostite c) sorbite d) bainite
 - vii. Gold has the crystal structure of- (a) BCC (b) FCC (c) HCP (d) RCC
 - viii. The name diffusion less transformation is associated with –
a) pearlite, b) bainite, c) martensite, d) cementite.
 - ix. Silicon when added to copper improves-
(a) machinability, (b) hardness, (c) hardness and strength, (d) strength and ductility
 - x. The ability of a material to be drawn into thin wire is known as-
a) Malleability b) Ductility c) Harness d) Brittleness
 - xi. The temperature (in °C) of peritectic reaction in iron-carbon phase diagram is-
a) 1539 b) 1493 c) 910 d) 1175
 - xii. The machine tool guide ways are usually hardened by
(a) vacuum hardening, (b) martempering, (c) induction hardening, (d) flame hardening
 - xiii. The hardenability is not affected by -
(a) air, (b) chemical composition of steel, (c) critical cooling rate, (d) quenching medium and method of quenching
 - xiv. To obtain martensite structure – a) slow cooling is to be done, b) quick cooling is to be done, c) air cooling is to be done, d) the cooling curve must cross C- curve.
 - xv. Stainless steel is so called because of its-
(a) High strength (b) High corrosion resistance
(c) High ductility (d) Brittleness

2. Fill in the blanks (any ten):

1x10

- a) The number of effective atoms in a HCP unit cell is _____.
- b) The ultimate stress of a material is _____ than the breaking stress.
- c) Plastic cables are made by _____ process.
- d) _____ metal has the highest electrical conductivity.
- e) Curie temperature for iron is _____.
- f) The iron ore mostly used for the production of pig iron is _____.
- g) Carbon percentage in cast iron is _____.
- h) Most metals has _____ structure.
- i) Y-alloy is an important alloy of _____.
- j) The steel used for making shafts is _____.
- k) Babbit metal is a _____ based alloy.
- l) _____ as a medium is used for fastest cooling rate of steel quenching.
- m) The cupola is used to manufacture _____.
- n) BUNA- N is also called _____.
- o) The heat treatment process used for softening of hardened steel is _____.

3. Answer the following questions (any ten): <https://www.wbscteonline.com>

1x10

- a) Why polymer cannot be remoulded?
- b) What is the full form of PVC?
- c) What is creep?
- d) What is the last step in magnetic particle test method?
- e) What is 18:4:1 tool steel?
- f) Give two examples of Thermoplastics.
- g) Write two applications of Muntz metal.
- h) Give two examples of non destructive testing.
- i) Briefly state the application of Inver.
- j) What is the unit of strain?
- k) What is unit cell?
- l) What is meant by work hardening?
- m) What is the difference between bronze and brass?
- n) What does ASTM stand for?
- o) What is radiography test?

Group B

4. Answer the following questions (any six):

2x6

- i. What are the desirable properties of bearing materials?
- ii. How plain carbon steel is classified?
- iii. What are composite materials?
- iv. Write down the difference between hardness & toughness.
- v. What is Duralumin?
- vi. Write the composition of '40 Ni 2 Cr 1 Mo 25' steel.
- vii. What is an 'Allotropic metal'?
- viii. What is powder metallurgy process?
- ix. What is the purpose of annealing?
- x. What is phase diagram?

Group C

5. Answer the following question (any one):

1x6

- a) Draw the 'Iron-carbon equilibrium diagram' and explain it. Briefly describe Surface hardening and tempering as heat treatment processes.

4+2

- b) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of BCC crystal structure. 2+2+2
- c) What do you mean by a unit cell? Volume of a unit cell and one atom of the unit cell are 100 nm^3 and 10 nm^3 respectively. If the unit cell has 5 numbers of effective atoms then find out the atomic packing factor (APF) of the unit cell. 2+4
6. Answer the following question (any one): 1x6
- a) Explain the necessity of heat treatment for steel. Describe the process of quenching. 4+2
- b) Write short notes on - (i) induction hardening, (ii) flame hardening 3+3
- c) Describe normalizing process. State its purpose. 4+2
7. Answer the following question (any one): 1x6
- a) State two differences between thermosetting plastics and thermoplastics with proper examples. Give one application for each of the followings- (a) ABS, (b) Acrylics, (c) Bakelite. 3+3
- b) Differentiate between destructive testing and non destructive testing. Write short note on – i) Dry penetration test, ii) Magnaflux test. 2+2+2
- c) What is a polymer ? Give two examples. What is Vulcanized Rubber? 2+2+2

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MECHANICAL ENGINEERING MATERIALS*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): (1x20)
 - i) Coordination number FCC structure is - (a) 6 (b) 8 (c) 10 (d) 12.
 - ii) Pure iron at room temperature has the following crystal structure. - (a) BCC (b) FCC (c) HCP (d) SC.
 - iii) Atomic packing factor of HCP structure is - (a) 0.74 (b) 0.68 (c) 0.52 (d) none of these.
 - iv) The property by virtue of it a material can be rolled into thin sheet is called a) Ductility b) elasticity c) malleability d) Fatigue.
 - v) The continuous deformation under steady load is called - (a) creep (b) fatigue (c) corrosion (d) none of these.
 - vi) Hardest micro structure of steel is - (a) Ferrite (b) Martensite (c) Bainite (d) Pearlite.
 - vii) The steel having more than 0.8% carbon are called - (a) eutectoid steels (b) hypo eutectoid steel (c) hyper eutectoid steel (d) none of these.
 - viii) The property of material by virtue of which it undergoes permanent deformation with the application of a load, is called - (a) hardness b) plasticity c) Fatigue d) elasticity.
 - ix) In Annealing heat treatment cooling is done in (a) air (b) Furnace (c) water (d) None of these
 - x) Product of blast furnace is - (a) Wrought Iron (b) Cast Iron (c) Pig Iron (d) Steel
 - xi) In low carbon steel C % varies from - (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - xii) Cast iron is a - (a) ductile material b) malleable material c) brittle material d) tough material.
 - xiii) Which of the following iron exist at room temperature? - (a) alpha Iron (b) Beta Iron (c) gamma Iron (d) delta Iron
 - xiv) The hardness of steel increases if % of carbon (a) decreases (b) Increases (c) all of the above (d) None of the above
 - xv) Tempering heat treatment is done after - (a) Hardening, (b) normalising, (c) Annealing, (d) None of the above.
 - xvi) Which is surface hardening process? a) Annealing b) Normalising c) Nitriding d) hardening
 - xvii) The lower critical point for all steel is- (a) 600°C (b) 723°C (c) 700°C (d) 913°C.
 - xviii) Brass is an alloy of - (a) copper and zinc (b) copper and tin (c) copper, Aluminium (d) none of these.

- xix) Duralumin is an alloy of – a) Aluminium b) iron c) Nickel d) None of the above
- xx) Babbit metal is used in – (a) Bearing (b) Gear (c) Shaft (d) None of these
- xxi) Bottles from thermoplastic materials are made by – (a) Compression moulding b) Extrusion moulding c) Injection moulding d) blow moulding.
- xxii) Long plastic pipes are produced by – (a) Compression moulding b) extrusion moulding c) Injection moulding d) blow moulding.
- xxiii) Examples of Thermosetting plastics are –(a) Electric switch (b) Pressure cooker handle c) all of the above d) None of the above
- xxiv) Example of non destructive test is- (a) ultrasonic test (b) radiographic test (c) magnetic particle test (d) all of these.
- xxv) Non- destructive testing is used to determine a) Location of defects, b) Chemical composition, c) Corrosion of metal, d) All of these.

Group B

2. a) Write down the names of different types of crystal structures in metals. (2)
b) Write down the number atoms per unit cell, co ordination number, atomic packing factors and relation between sides of unit cell and atomic radius of simple cubic and BCC structure. (6)
3. a) Define hardness, Brittleness, ductility and Elasticity . (4)
b) Draw schematic diagram showing flow path for making steel and cast iron , wrought iron in different furnaces from iron ore . (4)
4. a) Draw the cooling curve for pure iron showing salient points . (2)
b) Draw Iron carbon equilibrium diagram up to 2 % carbon showing micro constituent present in (4)
c) Classify steel according to % of carbon . <https://www.wbscteonline.com> (2)
5. a) Explain the various purpose of heat treatment. (3)
b) Describe hardening heat treatment process and also write down its application. (5)
6. a) Write short note on any one i) Nitriding ii) Flame hardening iii) Austempering (4)
b) Draw the TTT diagram for eutectoid steel showing salient points. (4)
7. a) Write down the names of two copper based alloys. (2)
b) What are the properties of aluminium alloys? (2)
c) Describe briefly the properties and application of bearing metals. (4)
8. a) Compare Thermoplastics and Thermosetting plastic . (3)
b) Describe Compression moulding or Injection moulding process with the help of simple sketch. (5)
9. Write short notes on any two i) Fibre reinforced Composite Material ii) Ceramics iii) Bakelite
iv) refractories v) polyester (4+4)
10. a) Write down the names of different Non destructive testing methods. (2)
b) What are the differences between destructive and non destructive testing methods? (2)
c) Describe any one non destructive testing method with suitable sketch. (4)



DECEMBER 2024

MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from the rest.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (Any ten): 2x10=20

- i) An alloy of copper and zinc is known as- (a) Brass (b) Monel metal (c) Bronze (d) Hindalium.
- ✓ ii) German silver contains- (a) 70% Cu, 30% Zn (b) 60% Cu, 40% Zn (c) 40% Zn and 1% tin (d) 60.45% Cu, 35.2% Zn, 5.35% Ni.
- ✓ iii) When the austenitic steel (above 723°C) is cooled in air, then the resulting structure will be- (a) Fine pearlite (b) Martensite (c) Coarse pearlite (d) None of these.
- iv) Which alloying element is added to cast iron mainly with a view to mitigate the ill effects of sulphur?
(a) silicon (b) phosphorus (c) manganese (d) molybdenum.
- ✓ v) Which of the following is basically used in forging purpose? (a) Medium carbon steel (b) High speed steel (c) High carbon steel (d) Mild steel.
- ✓ vi) Which of the following alloy is mainly used for aircraft application? (a) Bronze (b) Babitt metal (c) Duralumin (d) None of these.
- vii) Fibre reinforced plastic is- (a) alloy (b) ceramic (c) composite (d) metal.
- viii) The property which is desirable in a material subjected to shock loading – (a) Ductility (b) Malleability (c) Hardness (d) Toughness.
- ix) Cast iron is a product of- (a) Cupola (b) Bessemer converter (c) Open hearth furnace (d) Blast furnace.
- x) Red hot hardness of an alloy steel can be improved by adding - (a) Tungsten (b) Cobalt (c) Manganese (d) Both (a) & (b).
- xi) Which of the following is the hardest? (a) Austenite (b) Pearlite (c) Bainite (d) Martensite.

- xii) Which of the following is/are an example of Thermo-setting plastic? (a) Melamine (b) Epoxies (c) Bakelite (d) All of these.
- xiii) Low cost camera lenses are made by - (a) Acrylics (b) Nylon (c) ABS (d) None of these.
- xiv) The smallest part of a crystal lattice is known as - (a) Lattice structure (b) Lattice point (c) Bravais lattice (d) Unit cell.
- xv) The lower critical temperature (in °C) in iron-carbon phase diagram is - (a) 1539 (b) 1143 (c) 910 (d) 723.

GROUP-B

- 2) (a) Why tempering is required in Heat Treatment operation?
(b) Explain Martempering process with suitable TTT diagram. 2+6
- 3) (a) Classify Plain carbon Steels along with their composition and applications.
(b) Write down the composition and use of wrought iron. 6+2
- 4) (a) Write down the advantages, limitations of Ultrasonic Crack Detection and its area of application. <https://www.wbscteonline.com>
(b) State the properties and application of ABS and polyesters. (4+4)
- 5) (a) What is the meaning of 'XT 72 W 18 Cr 4 V1'?
(b) Identify different alloying elements along with their effect on steel. (2+6)
- 6) (a) State the common properties of plastics.
(b) Explain briefly the working principle of Plastic Injection moulding process with neat sketch. (2+6)
- 7) (a) Describe Allotropy of Iron.
(b) Draw Iron-Carbon equilibrium diagram indicating different phases, their carbon percentage and temperature of transformation. (2+6)
- 8) (a) Write down the property, composition and use of Y-alloy and Gun metal.
(b) Describe Flame Hardening process. (5+3)
- 9) (a) Write down the Coordination Number and Atomic Packing Factor for SC, BCC and FCC.
(b) Specify two important methods to prevent the corrosion. (6+2)

MECHANICAL ENGINEERING MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20
- Which of the following material has maximum ductility- i) Mild steel ii) Copper iii) Nickel iv) Aluminium
 - ☒ An eutectoid steel consists of- i) wholly pearlite ii) wholly austenite ☒ iii) pearlite and ferrite iv) pearlite and cementite
 - ☒ The hardness of steel depends upon the- i) amount of cementite it contains ☒ ii) amount of carbon it contains iii) contents of alloying elements iv) method of manufacture of steel
 - ☒ Blast furnace is used to produce- i) pig iron ii) steel iii) wrought iron iv) cast iron
 - The strength is the ability of a material to resist- i) deformation under stress ii) externally applied forces with breakdown or yielding iii) fracture due to high impact loads iv) none of these
 - ☒ In induction hardening _____ is high. ☒ i) current ii) voltage iii) frequency iv) temperature
 - ☒ Which of the following is an amorphous material- i) Mica ii) Glass iii) Lead iv) Silver
 - ☒ The type of space lattice found in alpha-iron is- i) FCC ☒ ii) BCC iii) HCP iv) None of these
 - i) The toughness of a material _____ when it is heated. i) remains same ☒ ii) increase iii) decrease
 - j) Muntz metal (Yellow brass) contains i) 70% copper and 30% zinc ii) 60% copper ☒ and 40% zinc iii) 59% copper, 40% zinc and 1% tin iv) 60.45% copper, 35.2% zinc and 5.35% nickel
 - ☒ Vanadium when added to steel- i) decreases tensile strength ii) raises critical temperature iii) Sowers critical temperature ☒ iv) increases tensile strength
 - ☒ i) increases tensile strength ii) increases tensile strength iii) increases tensile strength ☒ iv) increases tensile strength
 - m) Which of the following property is desirable in parts subjected to shock and impact loads i) Strength ii) Brittleness iii) Stiffness iv) Toughness
 - n) The value of Atomic packing factor for BCC crystal structure is- i) 0.68 ii) 0.74 ☒ iii) 0.29 iv) none of these
 - ☒ o) Monel metal contains- i) 65% nickel, 15% chromium and 20% iron ii) 68% nickel, 29% copper ☒ and 3% other constituents iii) 80% nickel and 20% chromium iv) 80% nickel, 14% chromium and 6% iron

GROUP-B

2. ☒ a) Write short notes on - a) Bainite b) Spheroidite c) Martensite 6
☒ b) Write down few differences between Case Hardening & Flame Hardening 2
3. ☒ a) Write down composition of a) Munz metal b) Gun Metal c) Duralumin d) Y-Alloy 4
☒ b) What is Corrosion? Name few corrosion resistive materials? Mention two corrosion preventive methods. 4
4. ☒ a) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of FCC crystal structure. 2+2+2
☒ b) Describe the process of Nitriding. 2

- | | | | |
|----|----|--|-------|
| 5. | a) | Give one application for each of the followings - a)Acrylics b)Polyesters c)Epoxies d)Melamine | 4 |
| | b) | Briefly explain Dye penetration test. | 4 |
| 6. | a) | Describe Annealing process. State it's Purpose. | 4+2 |
| | b) | What is Martempering? | 2 |
| 7. | a) | Briefly explain a) Eutectic reaction b) Eutectoid reaction c) Peritectic reaction | 2+2+2 |
| | b) | Write down differences between White cast iron and Grey cast iron. | 2 |
| 8. | a) | Write down composition of Duralumin and Magnalium. | 2 |
| | b) | Write down one application each for laminated materials and fibre reinforced materials. | 2 |
| | c) | What do u mean by Vulcanization of Rubber? State its application. | 2+1 |
| 9. | a) | Write down differences between Hardness and Toughness. | 2 |
| | b) | What is powder metallurgy process? Write down few application of this process | 2+1 |
| | c) | Classify plain carbon steel and also mention their respective carbon percentage. // | 2+1 |

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MECHANICAL ENGINEERING MATERIALS

318/2(N)

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- i) ☒ Coordination number of FCC structure:
 a) 6 b) 8 c) 10 ☒ 12
- ii) ☒ Pure iron at room temperature has:
 a) BCC b) FCC c) HCP d) SC
- iii) ☒ Atomic packing factor (APF) of BCC:
 a) 0.74 ☒ 0.68 c) 0.52 d) None of these
- iv) ☒ Which property enables a material to resist scratching, cutting and abrasion:
 ☒ Hardness b) Plasticity c) Fatigue d) Elasticity
- v) ☒ Toughness of a material when heated:
 a) Remains same ☒ Increases c) Decreases d) none of these
- vi) ☒ Hardest microstructure of steel:
 a) Ferrite ☒ Martensite c) Bainite d) Pearlite
- vii) ☒ Hardness of steel increases when % C:
 a) Decreases ☒ Increases c) All of the above d) None of the above
- viii) ☒ An eutectoid steel consists of:
 ☒ Wholly pearlite b) Wholly austenite c) Pearlite & ferrite d) Pearlite & cementite
- ix) ☒ In the Austempering process, austenite changes into:
 a) Martensite b) Troostite c) Sorbite ☒ Bainite
- x) ☒ In annealing, cooling is done in:
 a) Air ☒ Furnace c) Water d) None of these
- xi) ☒ Tempering is done after:
 ☒ Hardening b) Normalising c) Annealing d) None of these
- xii) ☒ Lower critical point for steels (°C):
 a) 600 ☒ 723 c) 700 d) 913
- xiii) ☒ Babbitt metal is used in:
 ☒ Bearings b) Gears c) Shafts d) None of these
- xiv) ☒ Long plastic pipes are produced by:
 a) Compression moulding ☒ Extrusion moulding c) Injection moulding d) Blow moulding
- xv) ☒ NDT is used to determine:
 ☒ Location of defects b) Chemical composition c) Corrosion of metal d) All of these

GROUP-B

Answer any Five (05) questions.

- 2.(a) What do you mean by a unit cell? (2)
- (b) For a Body-Centered Cubic (BCC) crystal structure, determine: (2+4)
 - (i) Average number of atoms per unit cell
 - (ii) Atomic packing factor

3. Define the following mechanical properties: i) Elasticity ii) Ductility iii) Malleability iv) Fatigue (2+2+2+2)
4. Draw the Iron-Carbon phase diagram and label all important points. (8)
5. (a) Write a short note on Cementite and Pearlite. (3 + 3)
(b) Write the composition of High-Speed Steel (HSS). (2)
6. (a) Draw the TTT diagram for eutectoid steel showing all salient points. (5)
(b) Write short notes on Case hardening (3)
7. (a) Describe the Annealing process. (4)
(b) Write the purpose of Austempering and Martempering processes. (2 + 2)
8. (a) Write down the names of two copper-based alloys. (2)
(b) Write the composition of Muntz Metal and Gun Metal. (2 + 2)
(c) What are the applications of Monel Metal? (2)
9. (a) Compare Thermoplastics and Thermosetting plastics. (3)
(b) With the help of a diagram, describe the Injection Moulding process. (5)
10. (a) What are the differences between Destructive and Non-Destructive Testing? (3)
(b) Explain the Dye Penetration Test with a suitable sketch. (5)

